

Investigating Type II Supernovae with UV/opt/NIR Observations

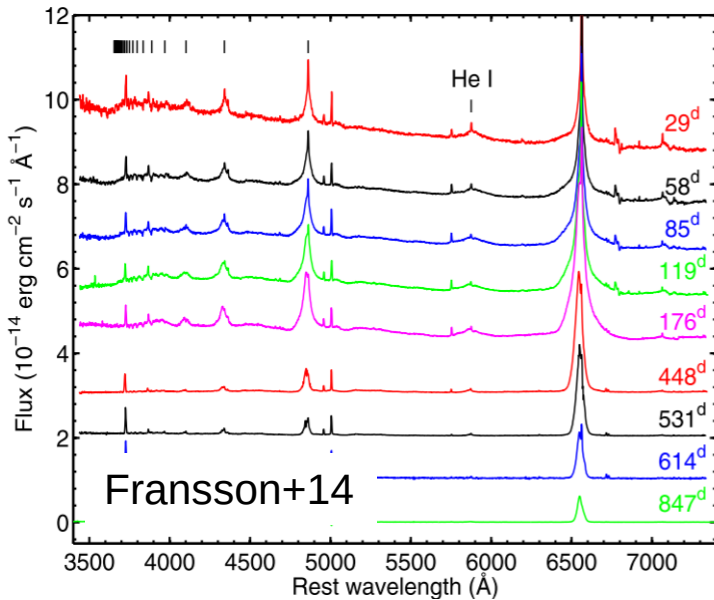
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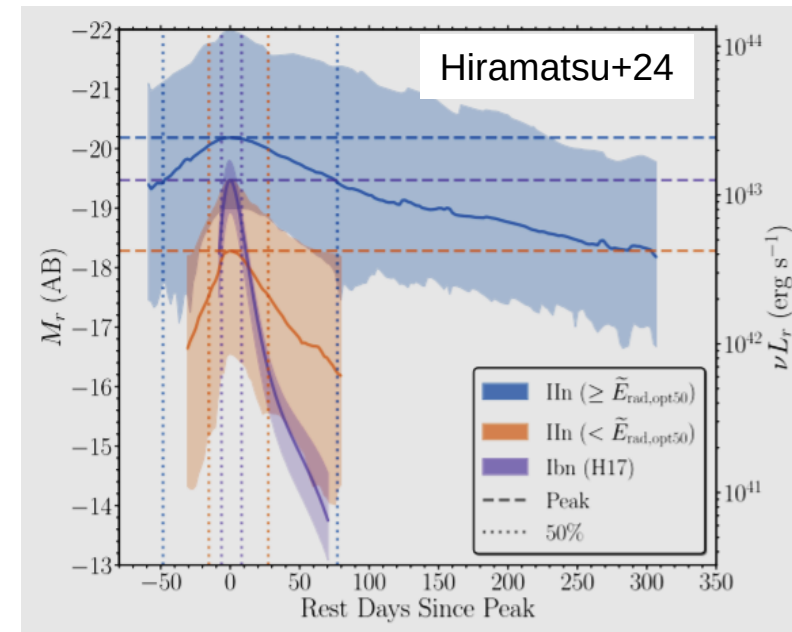
Outline of this talk

1. Type IIn supernovae
2. SN 2023vbg : The diversity of pre-SN activity
 - 2.1 2009ip-like SN IIn
 - 2.2 Light curves
 - 2.3 Black body fitting
 - 2.4 Spectral evolution
 - 2.5 Discussion
3. SN 2024dy : Shock-powered dust formation beacons
 - 3.1 Long-lived SN IIn
 - 3.2 Observation
 - 3.3 Light curves
 - 3.4 Comparison with long-lived type IIn SNe
 - 3.5 NIR excess in late phase
 - 3.6 Bolometric luminosity
 - 3.7 Spectral evolution
4. summary

1 Type IIn supernovae



- **Sign of H-rich CSM interaction**
 - blue continuum + narrow H α emission
- **Large diversity in LC**
 - luminous-slow and faint-fast (Hiramatsu+ 2024)



- **The origin of diversity ?**
 - CSM geometry, density profile, etc.
- **Mass-loss properties ?**
 - Duration, eruptive or steady



- UV/opt/NIR observations
- Follow-up from pre-supernova to late phase

SN 2023vbg :

The diversity of pre-SN activity

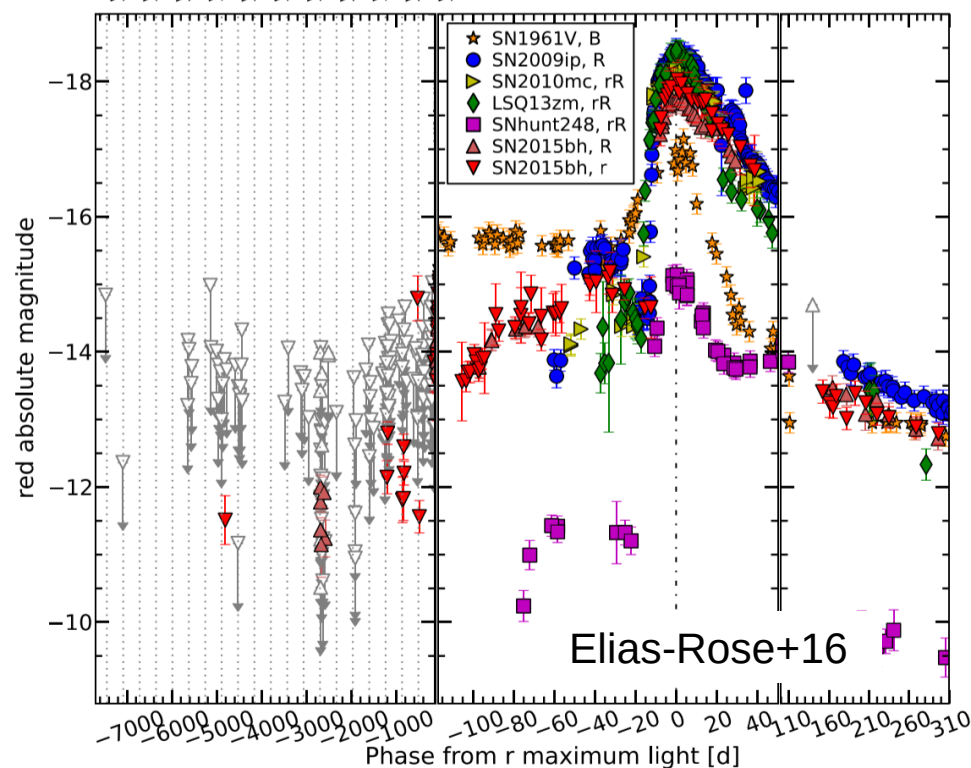
Goto et al. 2025 (under review)

[\[2504.15988\] SN 2023vbg: A Type II_n Supernova Resembling SN 2009ip, with a Long-Duration Precursor and Early-Time Bump](#)

2.1 2009ip-like SN IIn

2009ip-like SNe

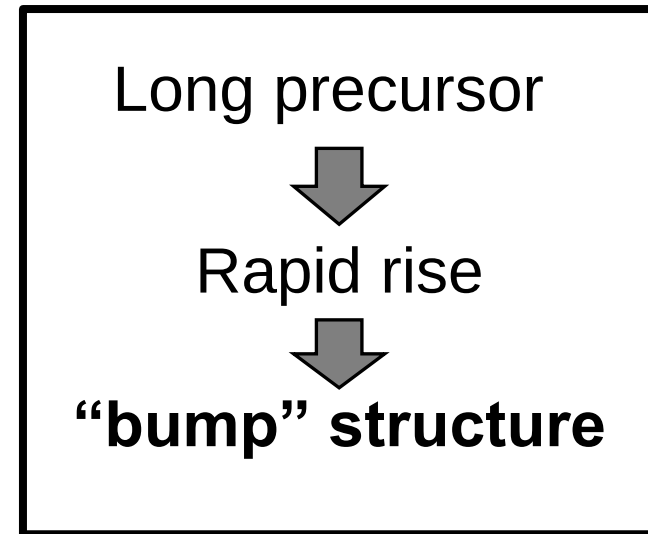
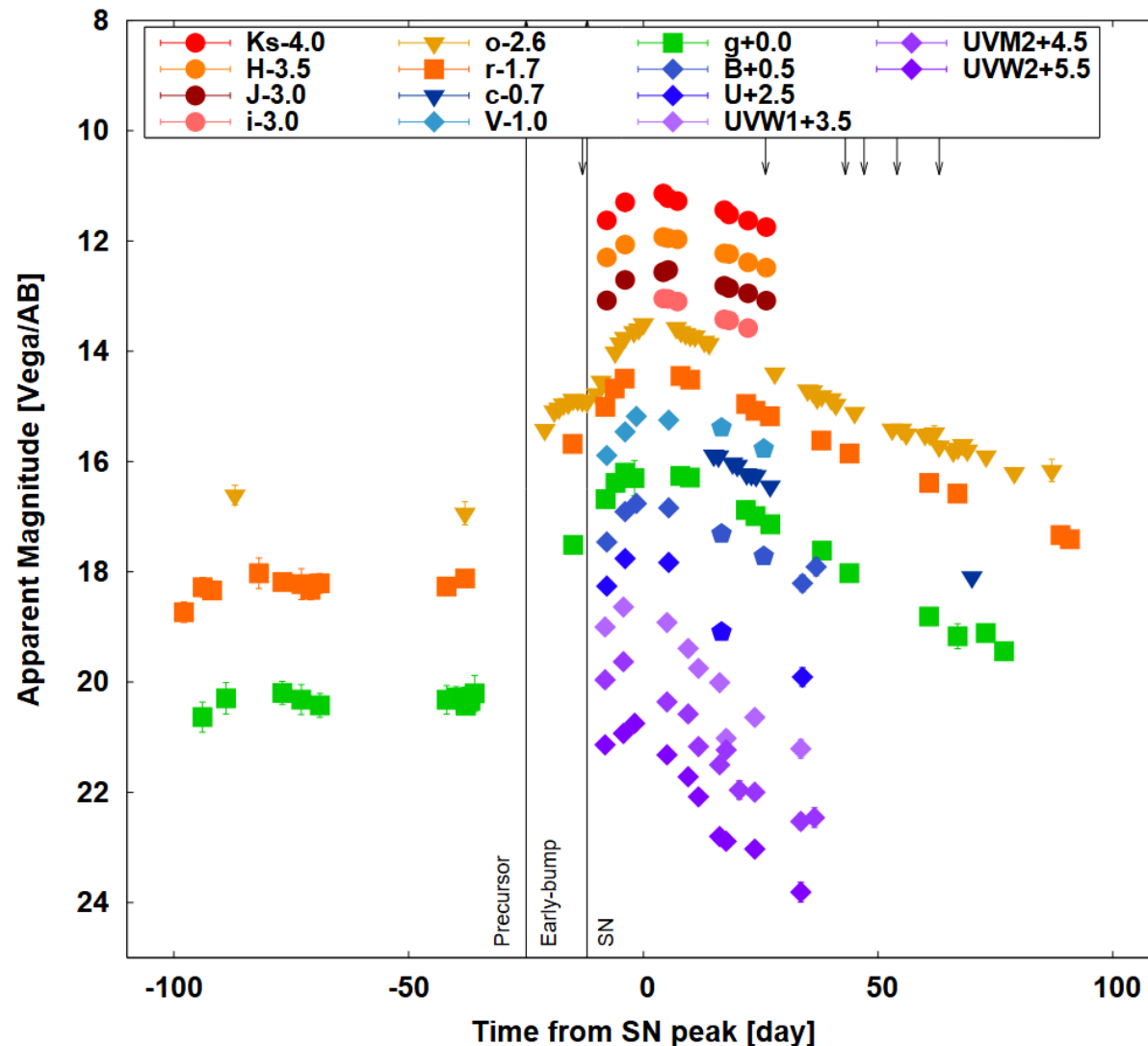
- Show months-to-years **precursor activity**
- Rise and fade rapidly with similar peak luminosities
- Display **Strong CSM interaction**



- **Explosion scenario**
 - Late phase observation
- **Progenitor evolution**
 - Precursor monitoring
- **Mass-loss history**
 - CSM density & geometry

2.2 Light curves

SN 2023vbg (09ip-like SN IIn)



We define 3 phases

Precursor : ~ 27 d
Early-bump : 27 d ~ 12 d
SN event : 12 d ~

2.4 Spectral evolution

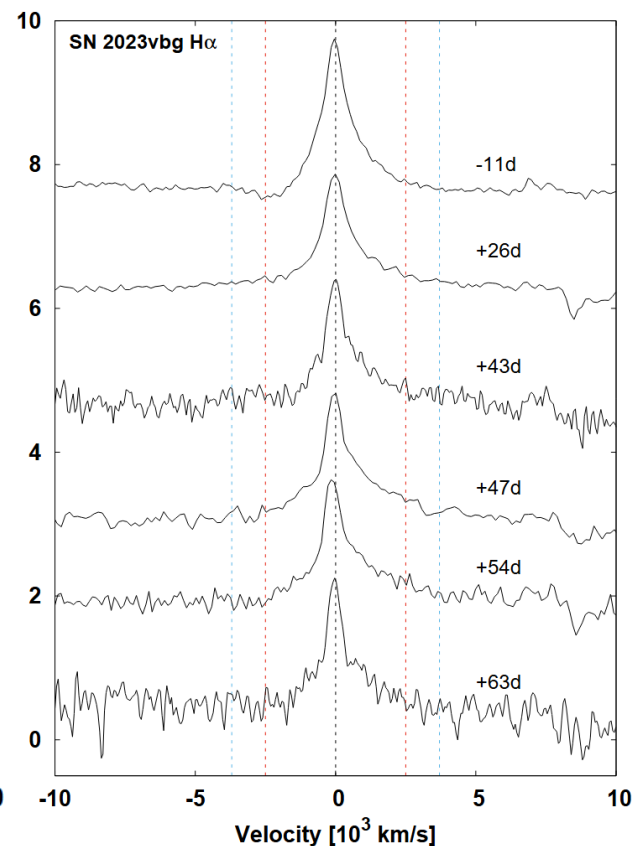
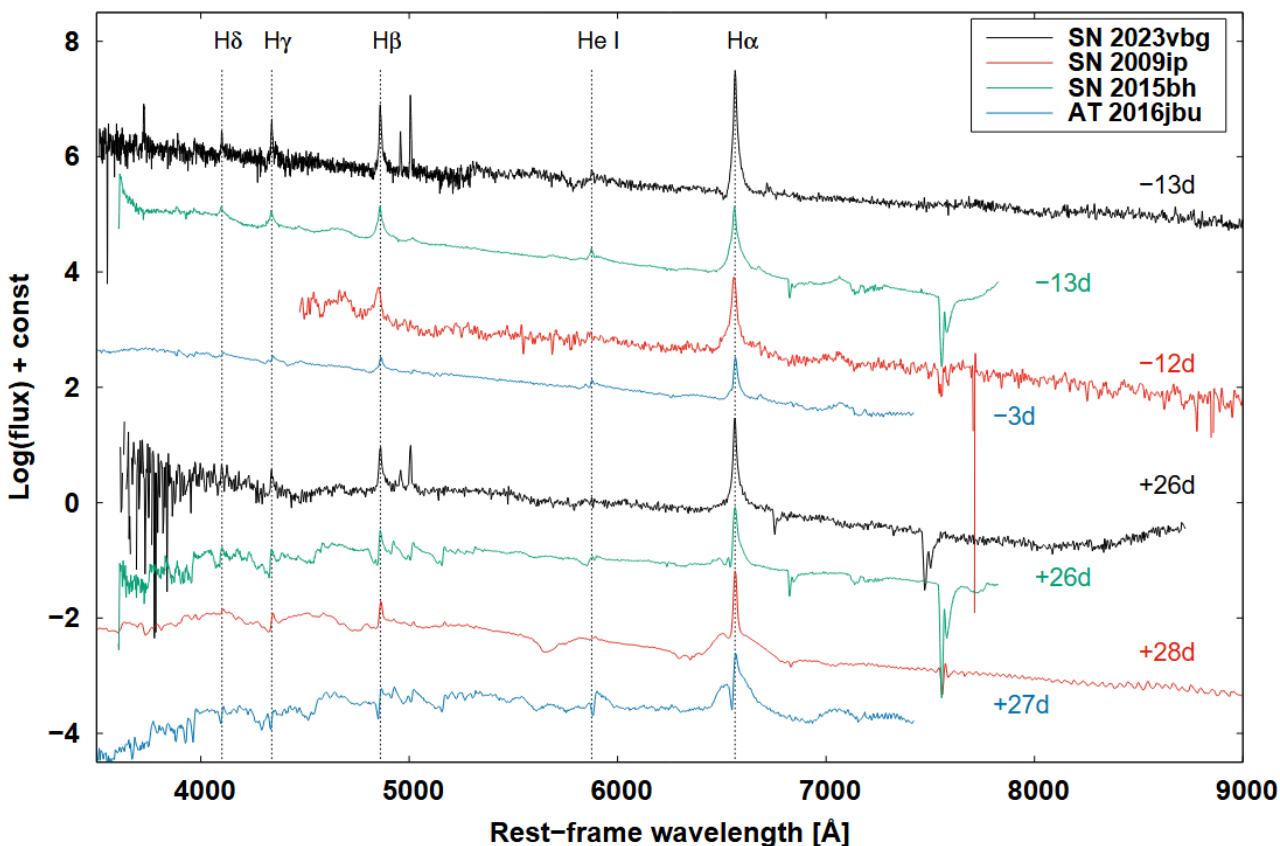
- Slow evolution, Balmer emission
- No other feature is seen

H α : **Narrow** + **Intermediate**

~ 500 km/s

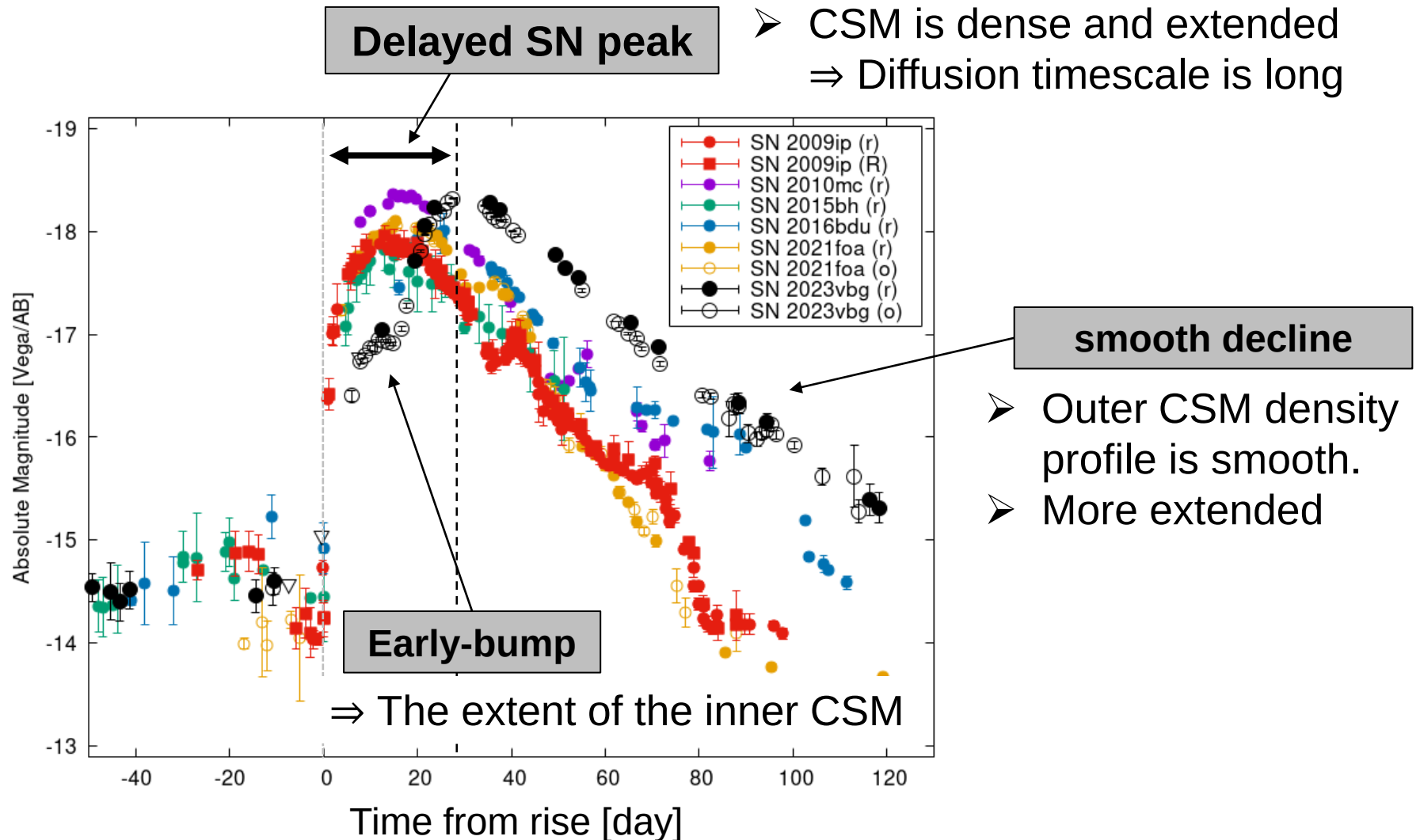
~ 3000 km/s

Unlike other 09ip-like objects,
no broad component is observed

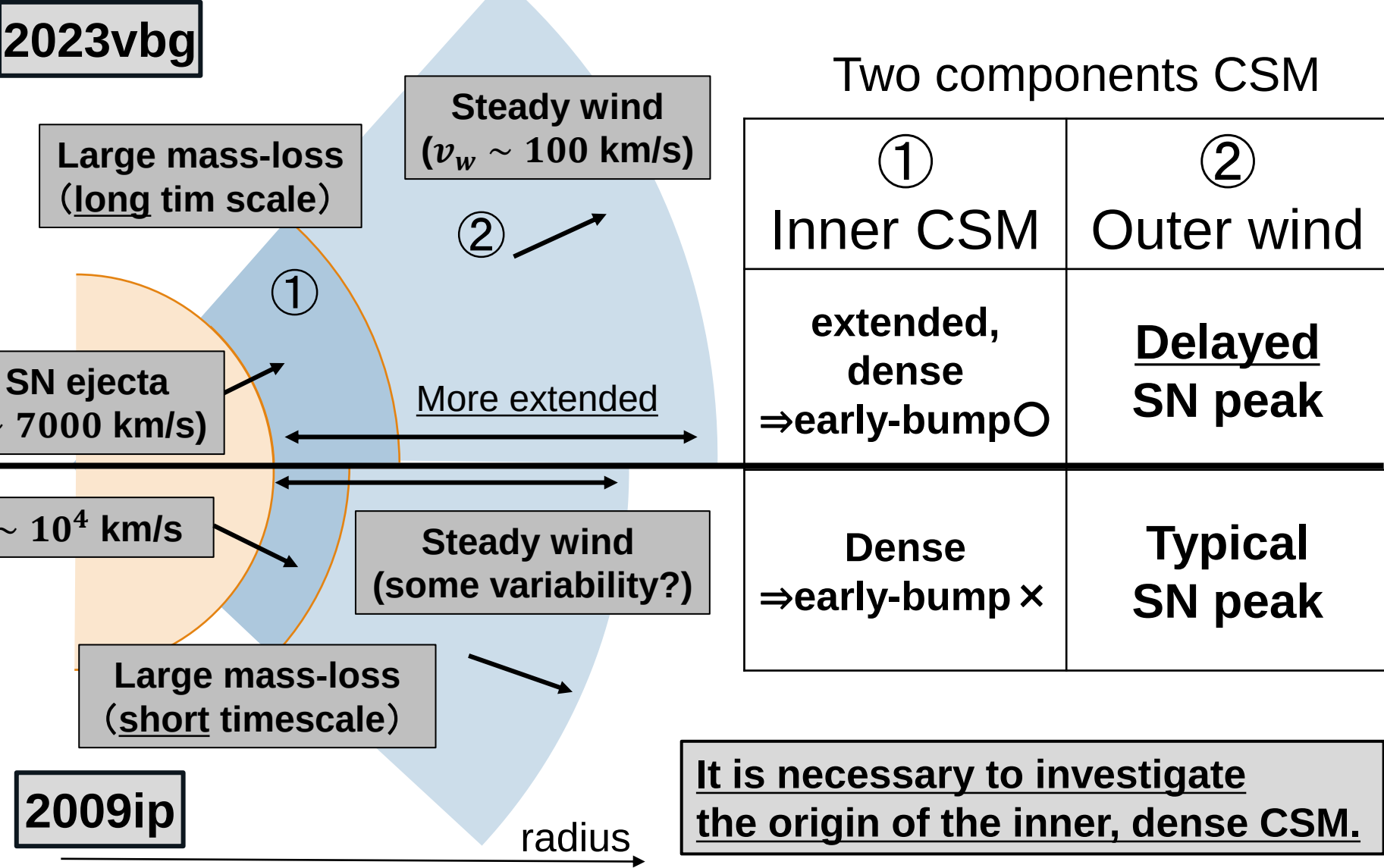


2.5 Discussion : Delayed SN peak

- comparison with $t=0$ aligned at the onset of the rise.



2.6 Discussion : Origin of early-bump

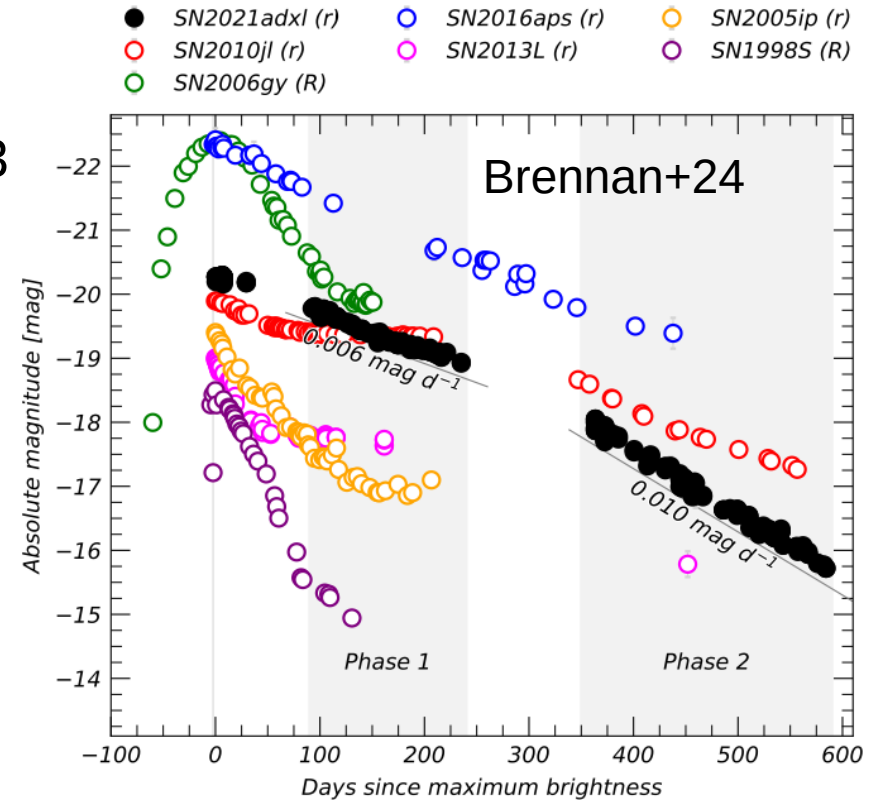
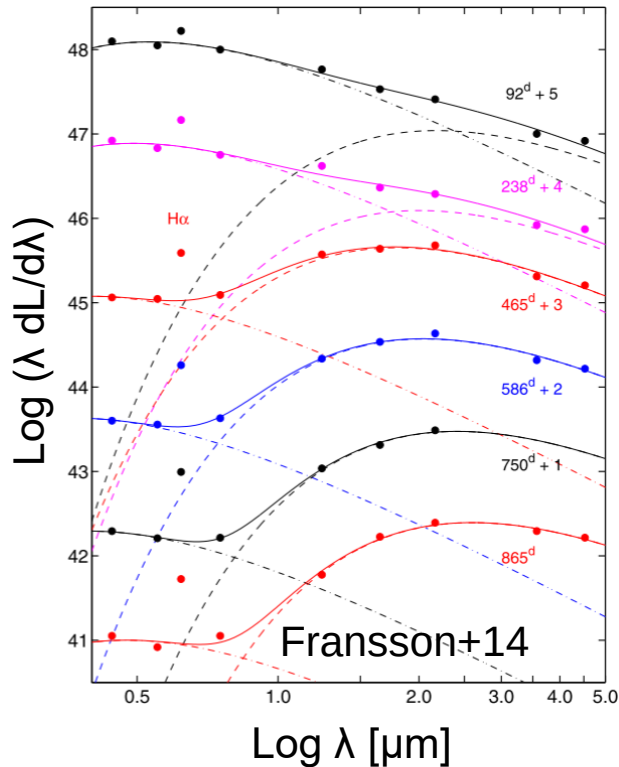


SN 2024dy : Shock-powered dust formation beacons

3.1 Long-lived SN IIn

➤ long-lived SNe

- Peak absolute magnitude : -18 to -23
- Slow light curve evolution
- Persistent strong CSM interaction
- **Early dust formation**
(e.g., SN 2010jl; Fransson+14)

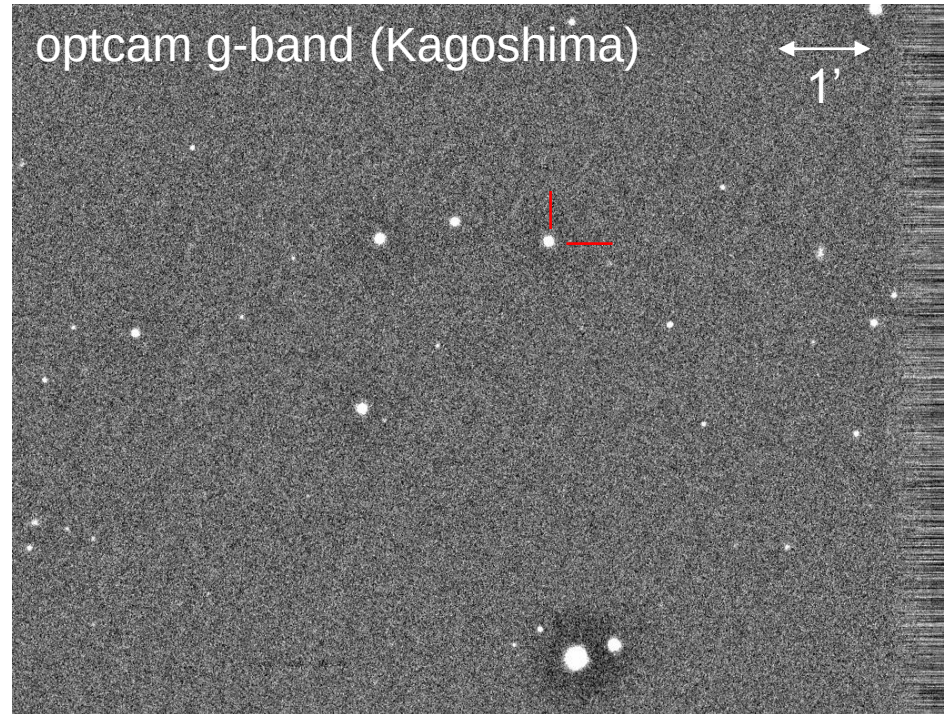


- **Energy budget**
- **Dust formation**
 - Grain size, distribution, etc.
- **Mass-loss history**
 - Rate, duration, geometry

3.2 Observations

SN 2024dy

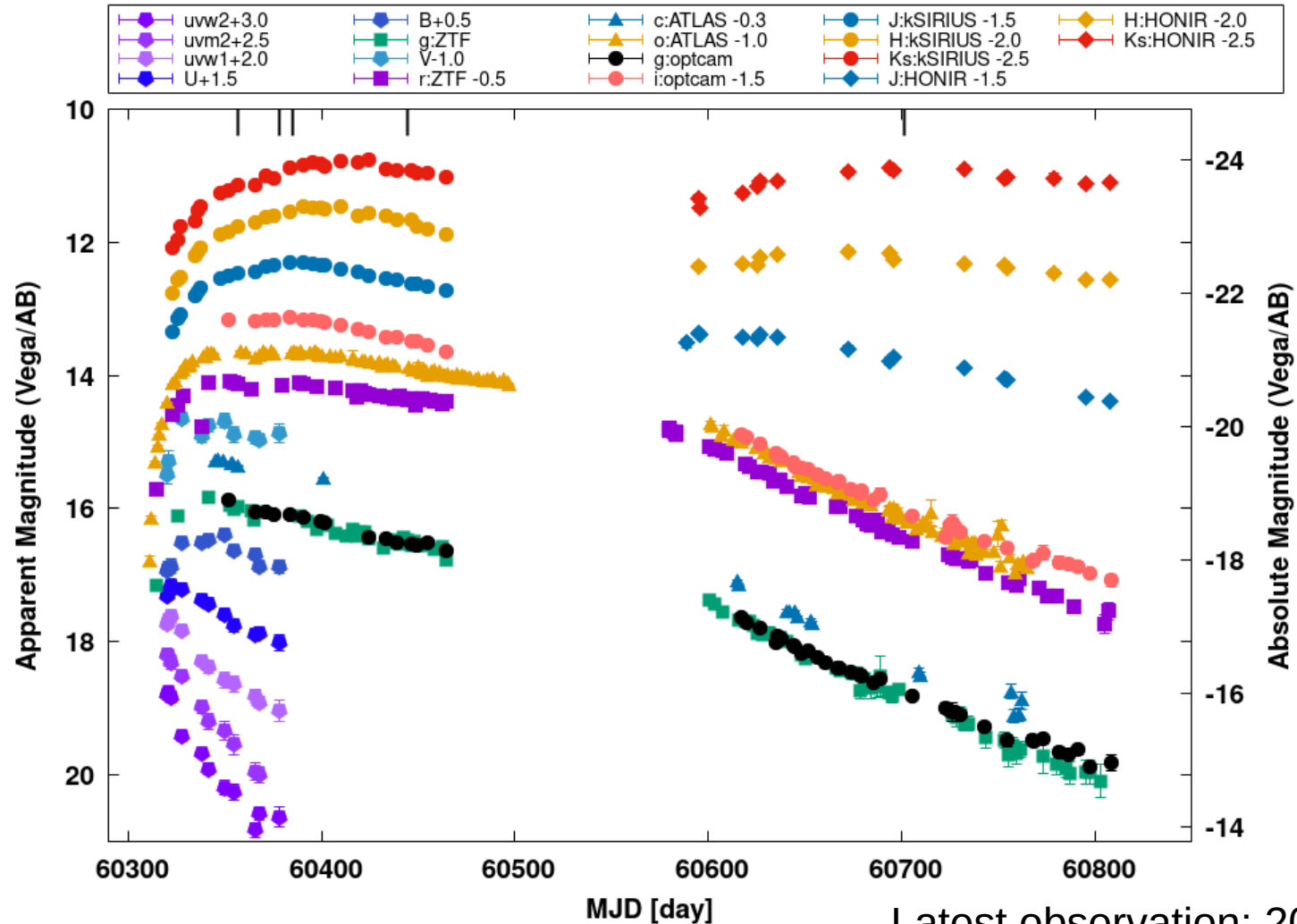
- ✓ Type IIn supernova
 - Narrow Balmer line
 - He lines are also seen
- ✓ Discovery date
 - 2024 January 3
- ✓ Redshift
 - $z = 0.021$
- ✓ The host galaxy is faint
 - ⇒ Contamination is negligible



- ✓ We conducted follow-up observations in optical and NIR wavelength.
 - **kSIRIUS + gi camera, Kagoshima (*gi*+*JHKs*)**
 - HONIR, Hiroshima (*JHKs*)
 - ADFOSC, ARIES (optical spec)
 - HFOSC, IIA (optical spec)

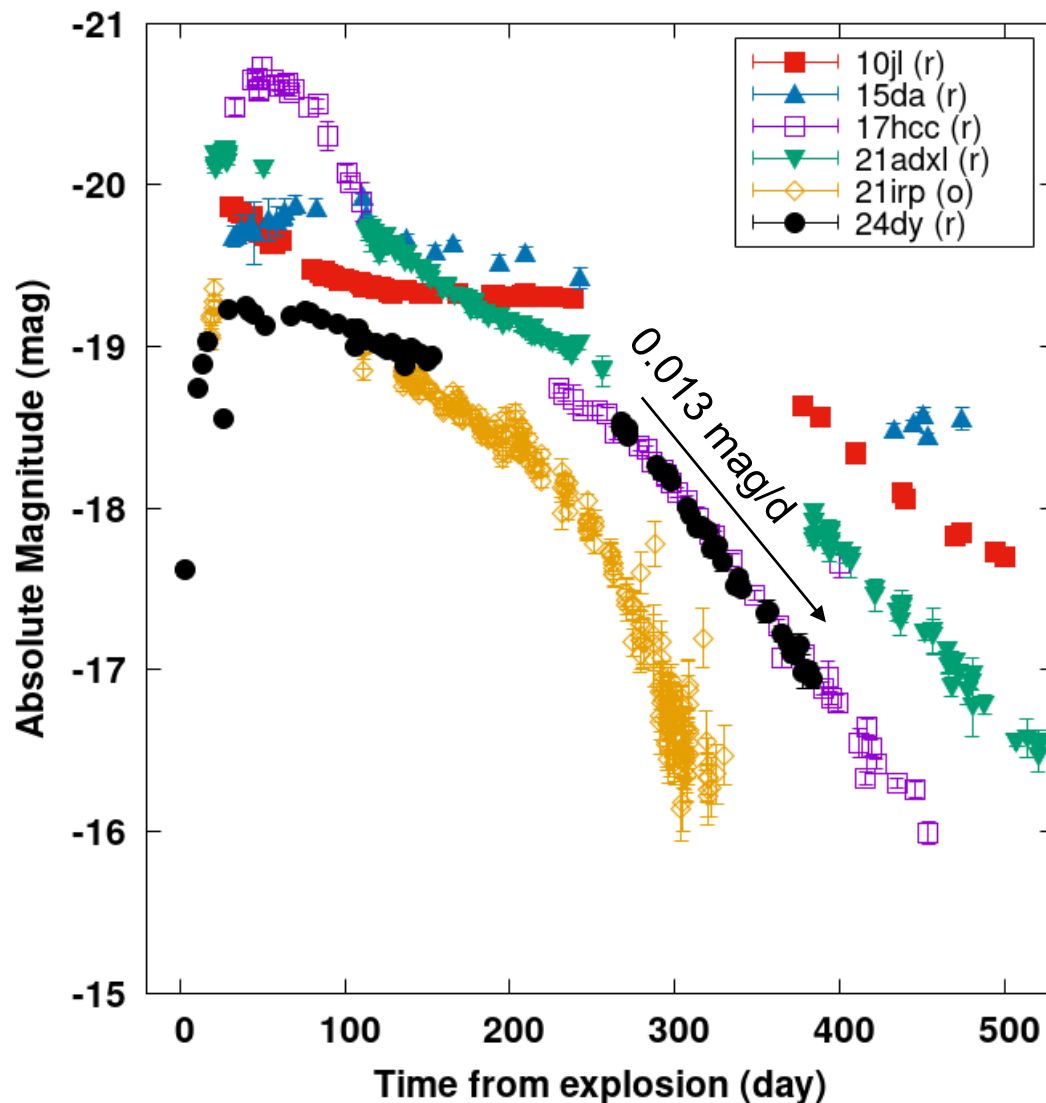
3.3 Light curves

Observation is ongoing ...



Archive data : Swift/UVOT (UV, *UBV*), ATLAS (*co*), ZTF (*gr*)

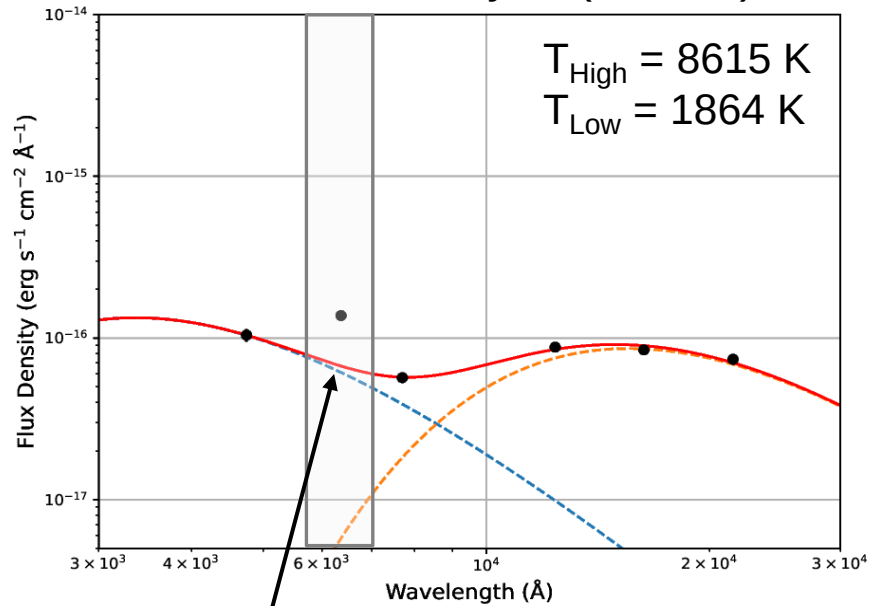
3.4 Comparison with long-lived type IIIn SNe



- Before solar conjunction
⇒ Fast rise + slow decline
- After solar conjunction
⇒ Faster decline
- Peak magnitude ≈ -19.2 mag
- Lower edge of this group
- Decline rate ~ 0.013 mag/d
- **Similar to 2017hcc**

3.5 NIR excess in late phase

Two blackbody fit ($t=421\text{d}$)

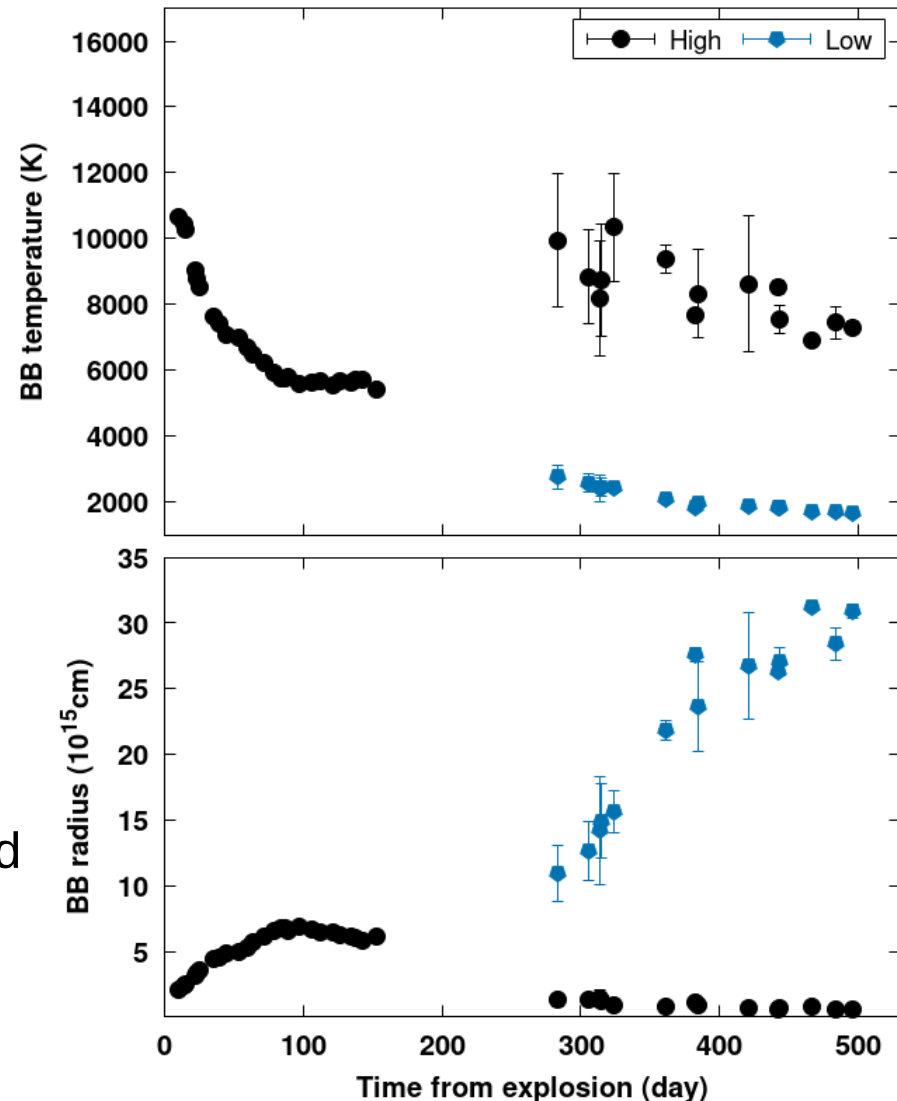


Excluded from the fitting
because of strong H α emission

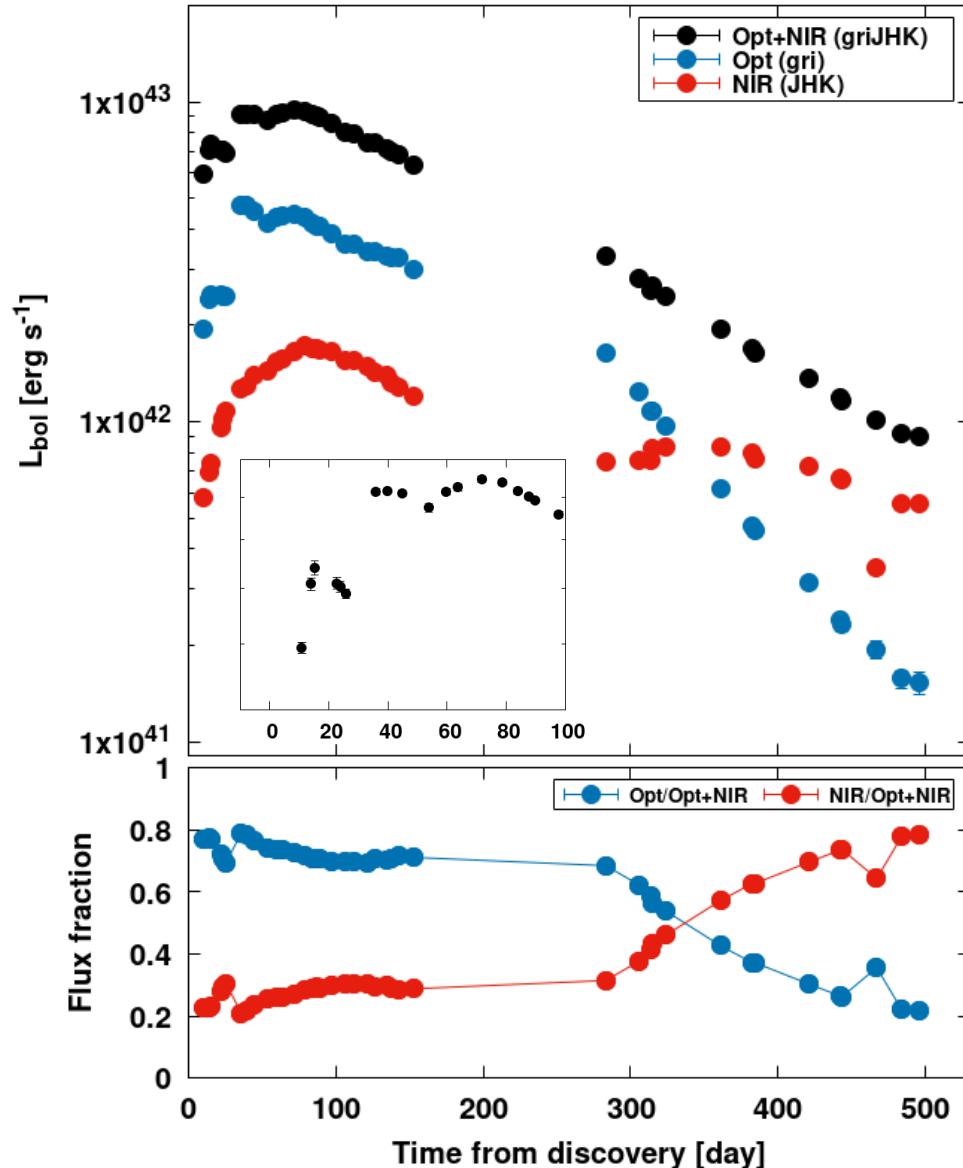
➤ NIR excess became apparent at $t \geq 280\text{d}$



Contribution from dust emission



3.6 Bolometric luminosity

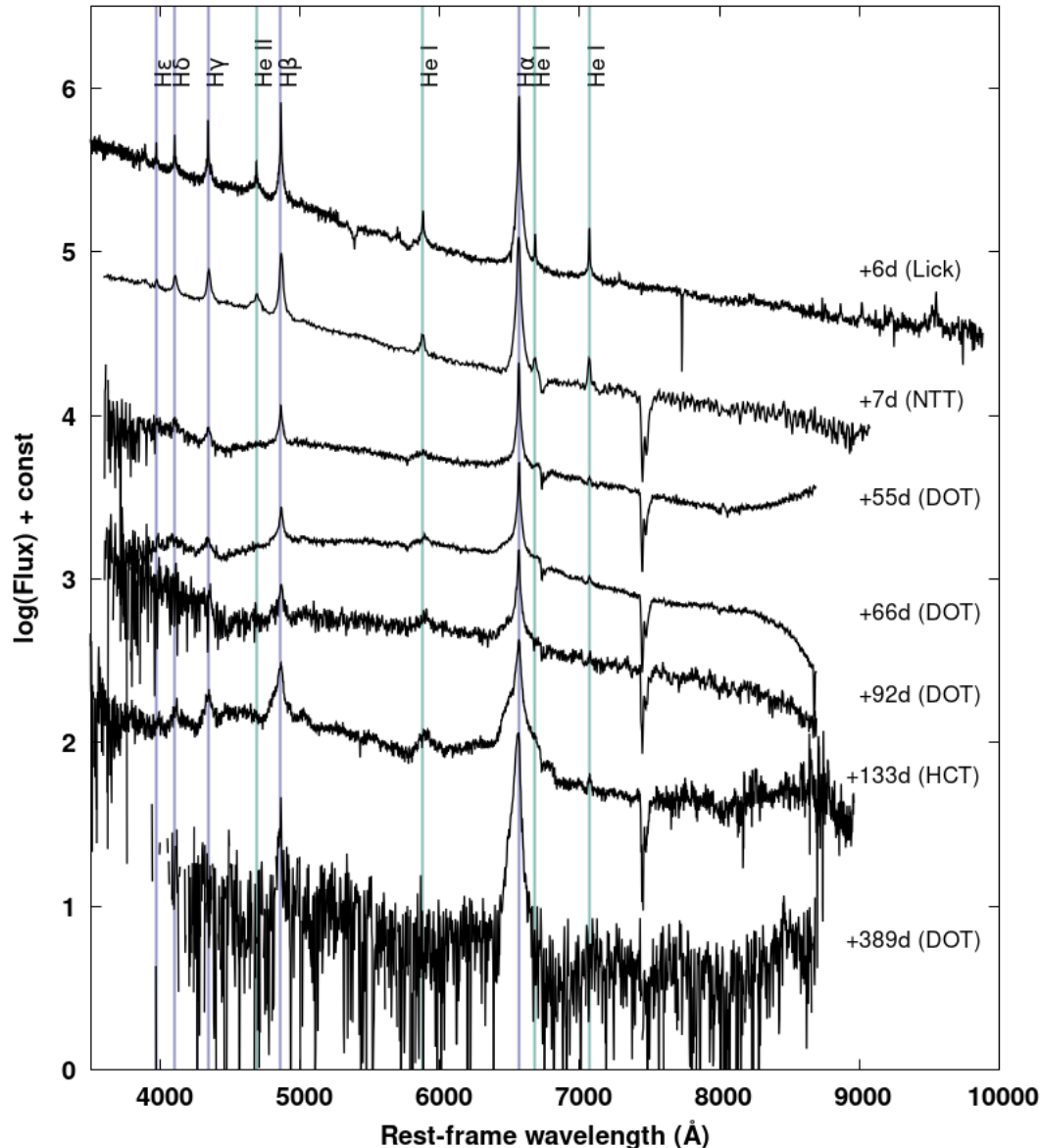


- Rise time ≈ 70 d
- NIR peak is late
 - rise time ≈ 100 d
- After 300 d
 - optical : steeper decline
 - NIR : flat
 ⇒ **NIR fraction is increasing**
- Mass-loss rate
 (Assuming $v_{\text{wind}} \approx 100 \text{ km/s}$)
 ⇒ **$0.012 \text{ M}_{\odot} \text{yr}^{-1}$**
 (using relation of Chugai & Danziger 1994)

For comparison

- 10jl (Fransson+14) : $\geq 0.8 \text{ M}_{\odot} \text{yr}^{-1}$

3.7 Spectrum evolution



We have 5 (+22) optical spectra

- 3.6m Devasthal Optical Telescope (DOT, ARIES)
 - 2.0m Himalayan Chandra Telescope (HCT, IIA)
- (The top two are taken from TNS)

Sign of CSM-interaction

- Blue continuum
- Narrow Balmer emission



Broad component Asymmetric line profile

- Blue-shifted Balmer emission
- **Dust formation ??**
- Ca II triplet? (+133d ~)

4 Summary

- SN 2023vbg
 - Shared several observational features with 09ip-like SNe.
 - **A unique “early bump”**
 - The diversity of CSM resulted in differences from 09ip-like SNe.
 - A key example for exploring the diversity of progenitor evolution
- SN 2024dy
 - Long-lived type IIn SN
 - After 300d, **showing NIR excess**
 - Spectra show strong CSM interaction
 - Observations and analysis are ongoing...
- **Follow-up observations with kSIRIUS will be continued.**